Clash Royale Deck Optimizer

Arena-weighted vulnerability analysis powered by real battle data & ML

Smart Analysis

6-factor vulnerability scoring

Game Feel

Role-aware card swaps

ML Pipeline

Top-player battle data

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The Problem

Clash Royale Deck Building is Hard

- 120+ cards to choose from
- 8-card decks must cover offense, defense, air, tanks, swarms & elixir cycle
- The meta changes constantly — what worked last season may not work now
- Most players guess based on YouTube videos or copying pro decks blindly

The Question

Can we build a system that analyzes a player's deck, understands the threats they face in their specific arena, and suggests targeted card swaps that actually make sense?

Not just "swap Knight for Graveyard" — but WHY, and for decks that play alike.

How It Works

1

Player enters tag

App fetches their current deck + last 25 battles from Supercell's official API

2

Analyze the meta

Battle log reveals what threats opponents actually run in this trophy range

3

Score the deck

6-factor vulnerability profile: air, tanks, swarms, spells, buildings, elixir

4

Optimize

Greedy search finds the single card swap that most improves the weighted score

5

Suggest

Up to 5 role-aware swap recommendations with explanations shown to the player

Vulnerability Analysis

Think of your deck like an investment portfolio — every card is an asset, every weakness is a risk factor

Air Exposure
Can you shoot down Balloon, Minions, Lava Hound?

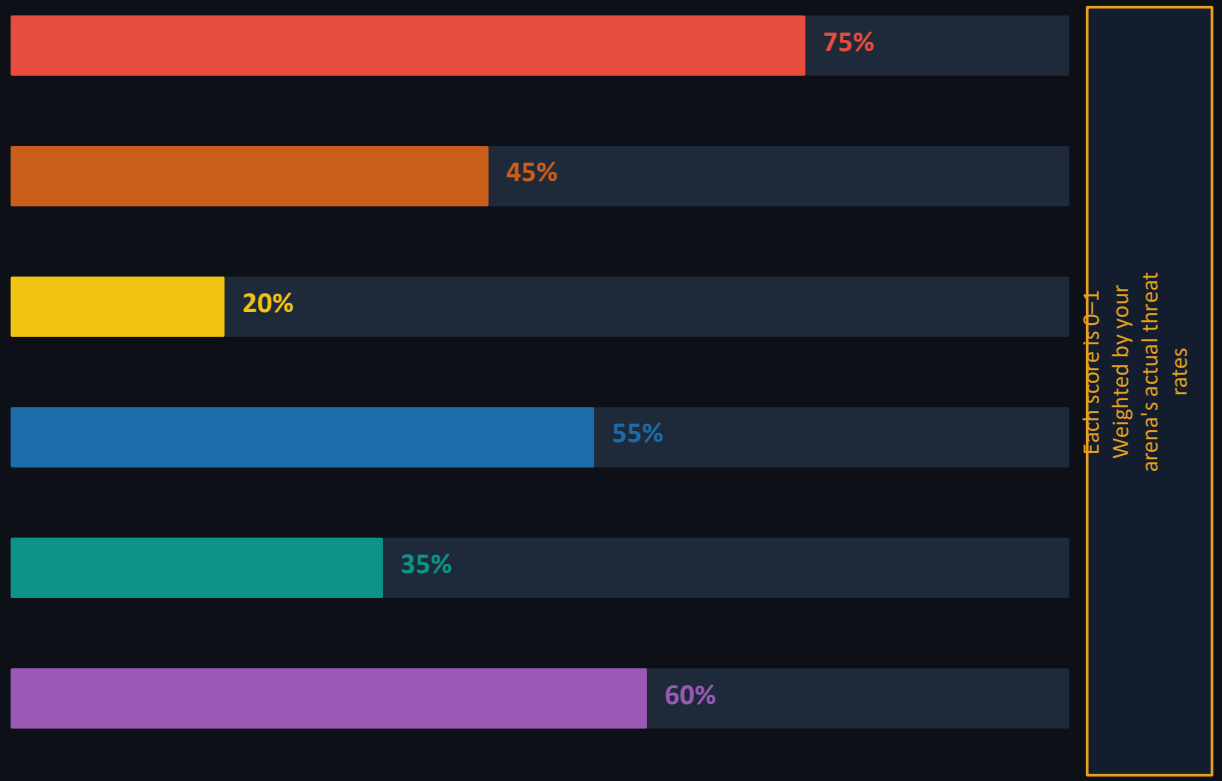
Tank Exposure
Do you have answers to Giant, Golem, PEKKA?

Swarm Exposure
Can you clear Skeleton Army, Goblin Gang?

Spell Exposure
Are your troops safe from Fireball, Poison, Log?

Building Exposure
Can you destroy defensive buildings (Tesla, Cannon)?

Elixir Risk
Is your deck too slow to cycle your win condition?



Arena Meta Weighting

Not all threats are equal in your arena

- At 3,000 trophies: Hog Rider is everywhere → air exposure matters more
- At 7,000 trophies: Golem beatdown dominates → tank exposure is critical
- We look at your last 25 battles and measure what % of opponents run each threat type
- Those rates become weights in the vulnerability score formula

The Formula

```
score = Σ (exposure_factor × threat_rate)
```

Air Exposure	60%	High
Tank Exposure	40%	Medium
Swarm Exposure	25%	Low
Spell Exposure	55%	High

Lower final score = better optimized deck for your arena

Adding Game Feel: Card Roles

✗ Before: Nonsensical Swap

Swap Knight → Graveyard
Both are "ground troops" statistically,
but they play completely differently!

✓ After: Role-Aware Swap

Swap Knight → Ice Golem
Both share: MINI_TANK + CYCLE roles
Valid replacement that preserves deck function!

14 Card Roles — Each card gets a frozenset of roles it fills:

WIN_CONDITION
Hog, Graveyard

MINI_TANK
Knight, Ice Golem

CYCLE
Skeletons, Ice Spirit

SWARM_CLEAR
Valkyrie, Arrows

ANTI_AIR
Mega Minion, Tesla

TANK_KILLER
Inferno Tower, PEKKA

SUPPORT_TROOP
Musketeer, Witch

BUILDING
Tesla, Cannon

Battle Data: Co-occurrence Confidence

Roles tell us IF a swap is valid. Battle data tells us WHICH valid swap is most likely to win.

How It Works

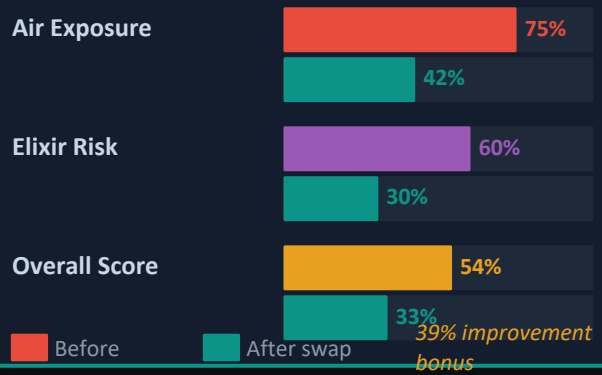
- Filter your last 25 battles to wins only
- For each winning deck, record which 8 cards appeared together
- Build a co-occurrence matrix: how often does Card B appear when Card A wins?
- High co-occurrence = high affinity score
- Affinity bonus (up to +35%) boosts role-valid swaps that fit your winning pattern

Real Example: Hog Rider Deck

Knight OUT → Ice Golem IN

✓ Shared roles: MINI_TANK + CYCLE → swap is valid

Vulnerability score improvement:



39% improvement + battle-proven pairing bonus

Next: ML-Powered Card Embeddings

Just like Word2Vec learns word meaning from sentences, we train it on winning decks to learn card meaning from co-occurrence

Top 1000 Players

get_location_rankings() fetches global leaderboard

Battle Logs Collection

fetch 25 battles per player
= ~25,000 battles total

Filter Winning Decks

Extract 8-card decks from wins only → training corpus

Word2Vec Training

Each deck = a sentence
Each card = a word

Card Embeddings

Similar cards cluster together in vector space

The Word2Vec Analogy

```
"king" - "man" + "woman" = "queen"
"Hog Rider" - "Knight" + "Ice Golem" ≈
valid swap
```

What This Enables

Cards used in similar ways end up close together in vector space — giving the optimizer data-driven game feel from the best players in the world

Tech Stack

Backend

- FastAPI — async Python web framework
- httpx — async HTTP client for API calls
- SQLAlchemy — database ORM
- PostgreSQL — user data & snapshots
- JWT — authentication tokens

Core Engine

- Custom vulnerability analyzer
- Greedy local search optimizer
- Arena meta weighter
- Role registry (14 card roles)
- Co-occurrence confidence scoring

Data & ML

- Supercell Official API
- RoyaleAPI proxy (static IP)
- RoyaleAPI JSON datasets
- Word2Vec card embeddings (planned)
- gensim — ML training library

Frontend & Deploy

- Single HTML file (no framework)
- Vanilla JS + CSS
- Railway — cloud deployment
- GitHub — version control
- LibreOffice-free static serving

Results & What's Next

✓ What's Working

- Full deck analysis with 6 vulnerability scores
- Arena meta weighting from real battle logs
- Role-aware optimizer (Knight → Knight-like cards only)
- Battle co-occurrence affinity bonus
- User accounts, saved snapshots, favorites

→ What's Next

- Word2Vec card embeddings from top 1000 players
- Win-rate prediction model (logistic regression)
- Deck archetype detection (Hog Cycle, Golem Beatdown…)
- Card level awareness (only suggest owned cards)
- Deployment to Railway for public access

Key Insight: The same patterns that make a great Clash Royale deck also mirror quantitative finance — factor exposure, portfolio rebalancing, and meta-weighted risk scoring.